

What we're doing

Collect rollouts from Eric's tuned contacts-v1 1.5B model (eval loss 2.7566) at scale, and learn how fast we can generate them on TPU. Sample 1000 training targets (round-0, $L \leq 512$, ≥ 5 contacts); for each, generate 1000 rollouts; score every rollout's precision/recall/F1; save the best-recall and best-F1 rollouts verbatim for a future "train on high-accuracy rollouts" experiment.

Why

Groundwork for the real question: does fine-tuning on high-accuracy rollouts beat re-epoching the data? First we need the rollouts and a feel for the generation cost at scale. Builds on exp82 (rollout+resample is the best LM-only recipe) and exp89 (the vLLM/iris-TPU path).

How (inference)

vLLM on iris TPU v5p-8, bf16 model, `**tensor_parallel_size=4**` (all 4 chips).
rollout+resample recipe (each rollout a fresh document realization). Prefixes pre-built locally; the TPU worker is marifold-free and resumes on restart.

Calibration — throughput

On one v5p-8 (10 length-stratified targets × 1000 rollouts): tp=4 gives **~11.3k tok/s aggregate (~16–18k steady-state)**, ~1.8–2× over tp=1 (which left 3 of 4 chips idle). 0% of rollouts hit the token cap. resample ≈ nsample on accuracy, ~15% slower. Projected full 1M-rollout run: ~0.86 h on 8× v5p-8.

Calibration — accuracy

Per-rollout accuracy is low on average but **best-of-1000 $\gg$ mean**, and strongly length-dependent: small proteins ($L \approx 30-80$) reach best recall 0.8–1.0, large ones ($L > 250$) stay near 0.15–0.2. This is the point of sampling many rollouts — the best one is far better than the typical one.

Full run — scale

1,000,000 rollouts (1000 targets $\times$ 1000) in ****~41 min wall on 8 $\times$ v5p-8**** (~4.6 v5p-8-hours of generation; 283 M tokens; 17.2k tok/s effective; 0% truncated; 0 failures). 10 $\times$ more targets $\approx$ ~7.5 h on 8 $\times$ v5p-8 — affordable.

Full run — accuracy

Best-of-1000 per target (all $\text{sep} \geq 6$): recall mean **0.335** (max 1.0), F1 mean **0.340** (max 0.955) — $\sim 3\times$ the typical single rollout. Strongly length-dependent (best F1 0.52 at $L \leq 100 \rightarrow 0.20$ at $L > 400$); 20% of targets reach $F1 \geq 0.5$. Sampling many rollouts surfaces far higher-quality contact sets than any single decode — the raw material for the train-on-rollouts idea.

Selecting rollouts without labels

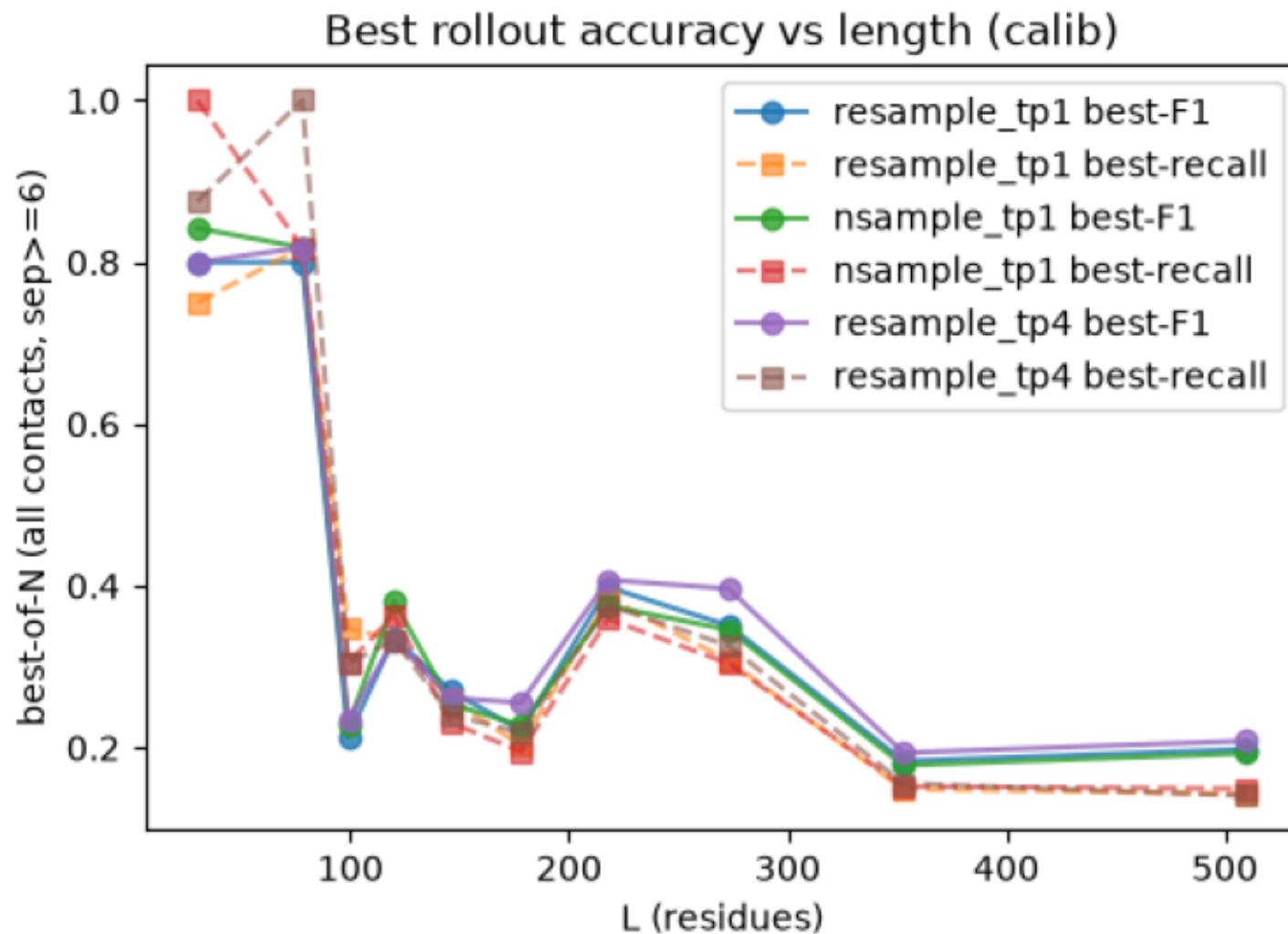
best-of-N needs ground truth. Label-free alternatives (F1, all $\text{sep} \geq 6$): **consensus voting** (aggregate contact occurrences across the 1000 rollouts, take top-K) = **0.256** — $2.2\times$ a random rollout, $\sim 75\%$ of the best-of-1000 oracle (0.341). Model **confidence** (NLL) is a weak selector (most-confident single rollout $0.13 \approx$ random 0.12) and **likelihood-weighting** the vote adds $\sim$ nothing at scale. Capturing NLL also costs $\sim 3.7\times$ the TPU time. Takeaway: use consensus, not confidence.

Deliverables

Per target: precision/recall/F1 + NLL + predicted contacts of all 1000 rollouts + the best-recall and best-F1 rollouts saved verbatim (complete contacts-v1 documents). Public HF bucket + interactive explorer: `explore_results.ipynb` (Colab).

calib_bestacc_vs_L.png

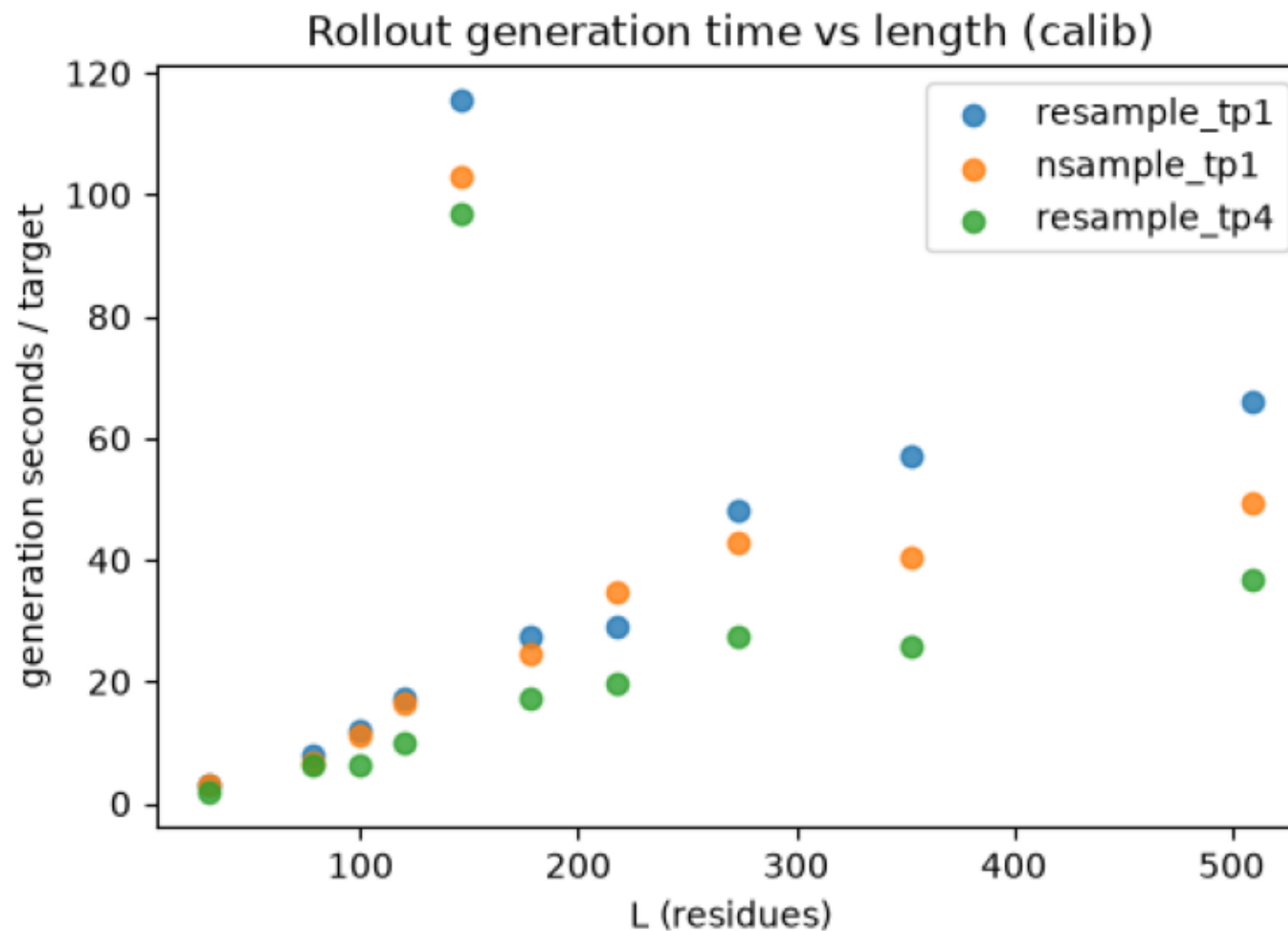
Best-of-N recall / F1 per target vs length.



```
$ aggregate_results.py --runs resample_tp1=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/calib_resample nsample_tp1=gs://marin-us-east5/protein-struct
```

calib_gentime_vs_L.png

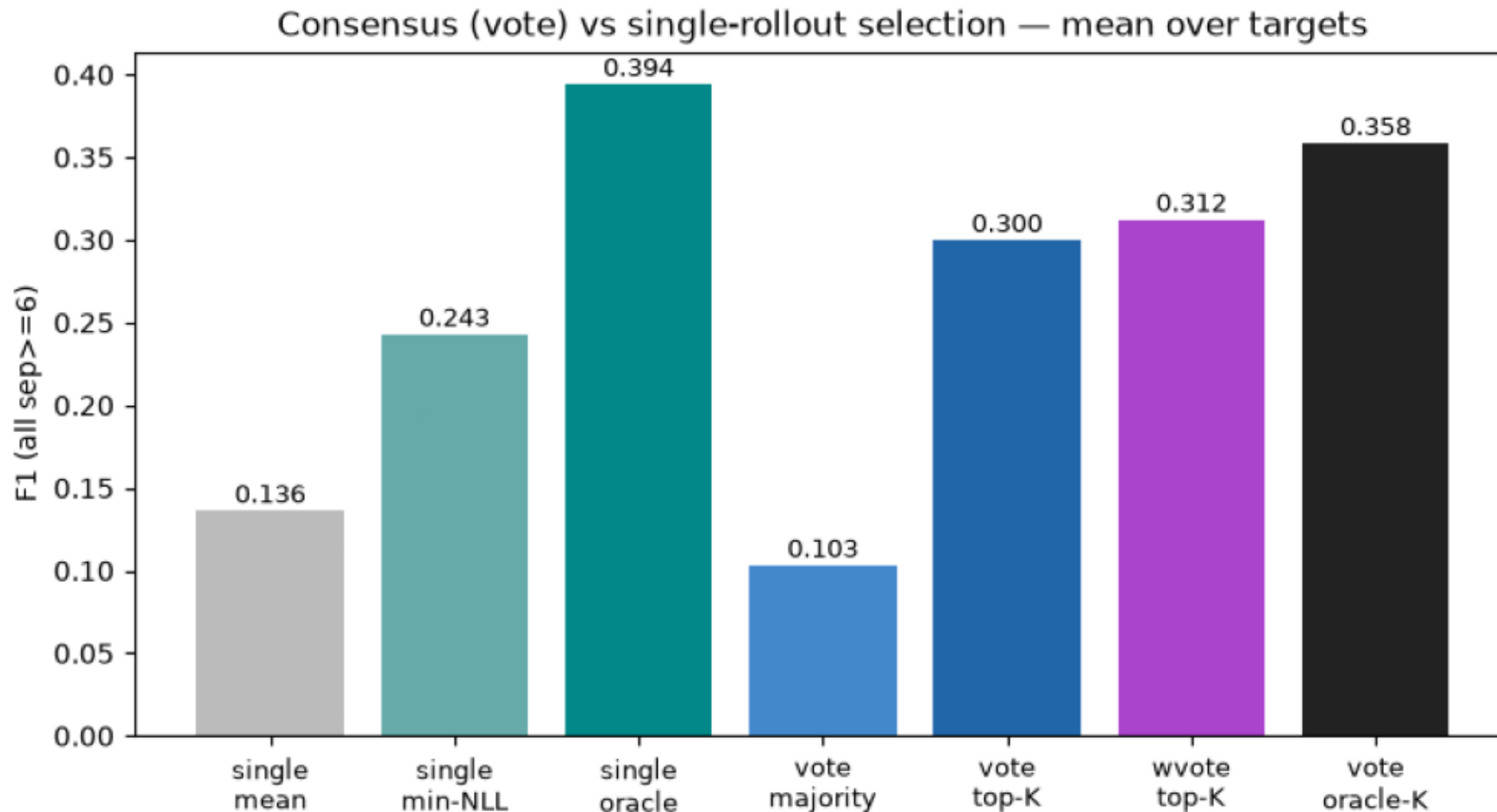
Per-target rollout generation wall time vs protein length.



\$ aggregate_results.py --runs resample_tp1=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/calib_resample nsample_tp1=gs://marin-us-east5/protein-struct

ens_calib_strategies.png

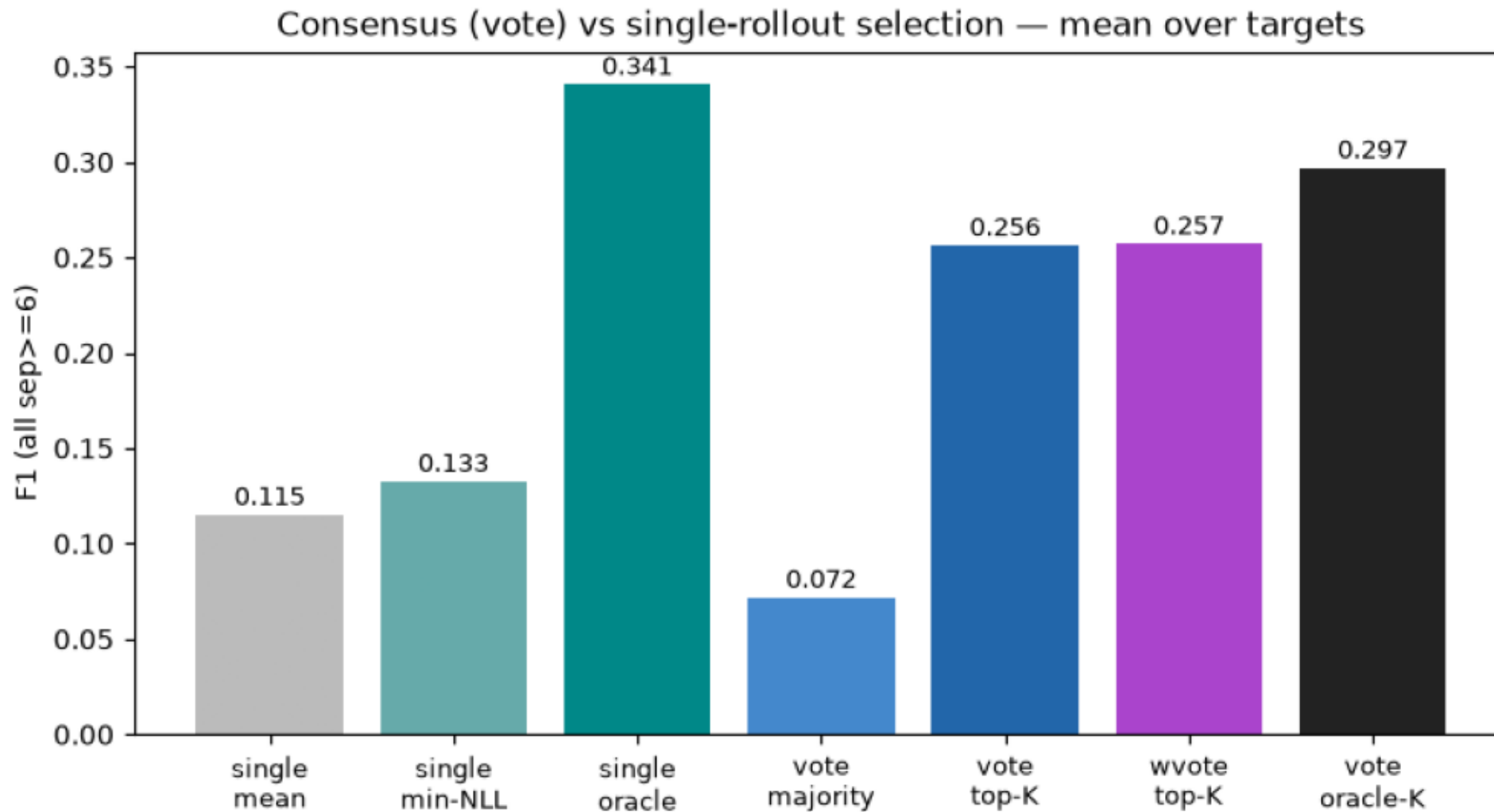
Per-target F1 by strategy. single-* pick one rollout; vote-* aggregate contact occurrences across 1000 rollouts (majority / top-K@mean-size = label-free; oracle-K needs GT). wvote weights rollouts by softmax(-NLL/token).



```
$ analyze_ensemble.py --run gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/calib_ens --targets data/calib_targets.parquet --temps 1.0,0.3 --out-prefix
```

ens_full_strategies.png

Per-target F1 by strategy. single-* pick one rollout; vote-* aggregate contact occurrences across 1000 rollouts (majority / top-K@mean-size = label-free; oracle-K needs GT). wvote weights rollouts by softmax(-NLL/token).

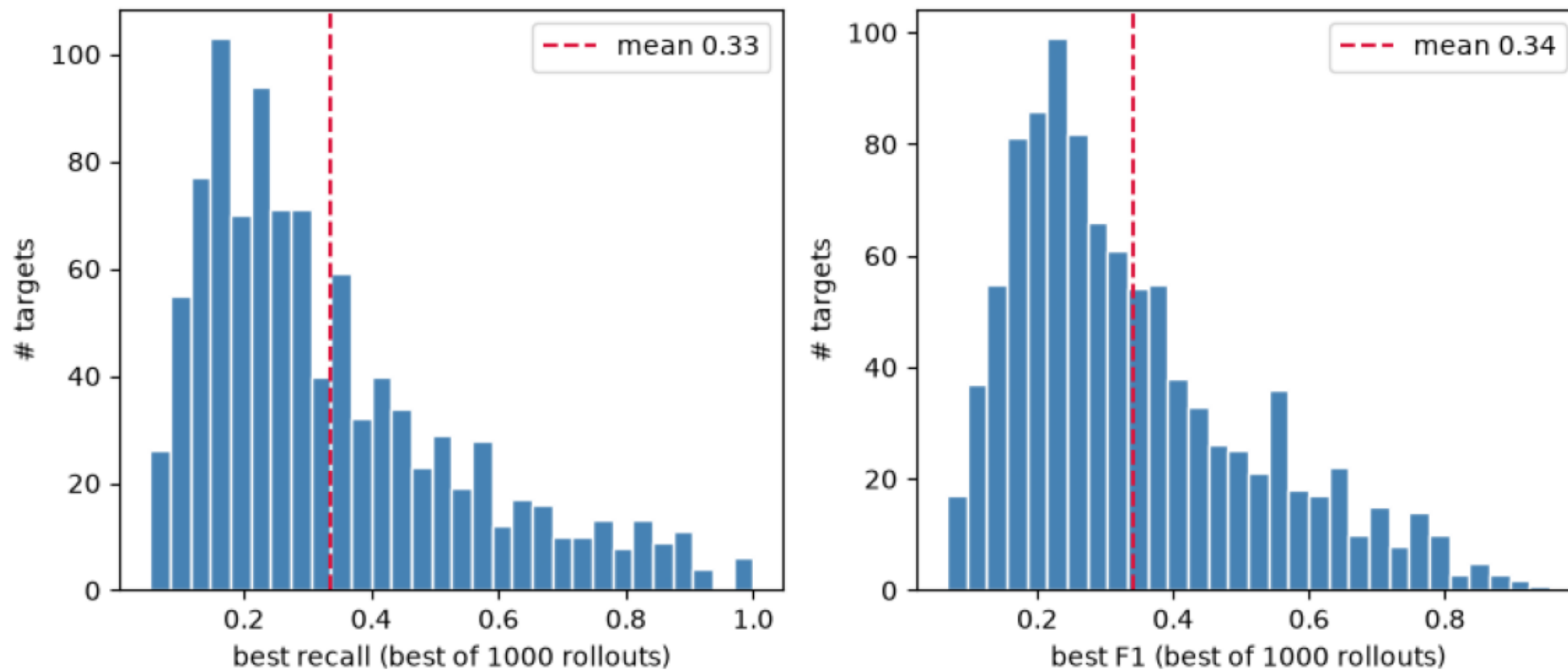


```
$ analyze_ensemble.py --run gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full_v2 --targets data/targets.parquet --temps 1.0,0.3 --out-prefix ens_full
```

full_best_hist.png

Distribution of per-target best-of-1000 recall and F1 (all contacts, sep $\geq$ 6).

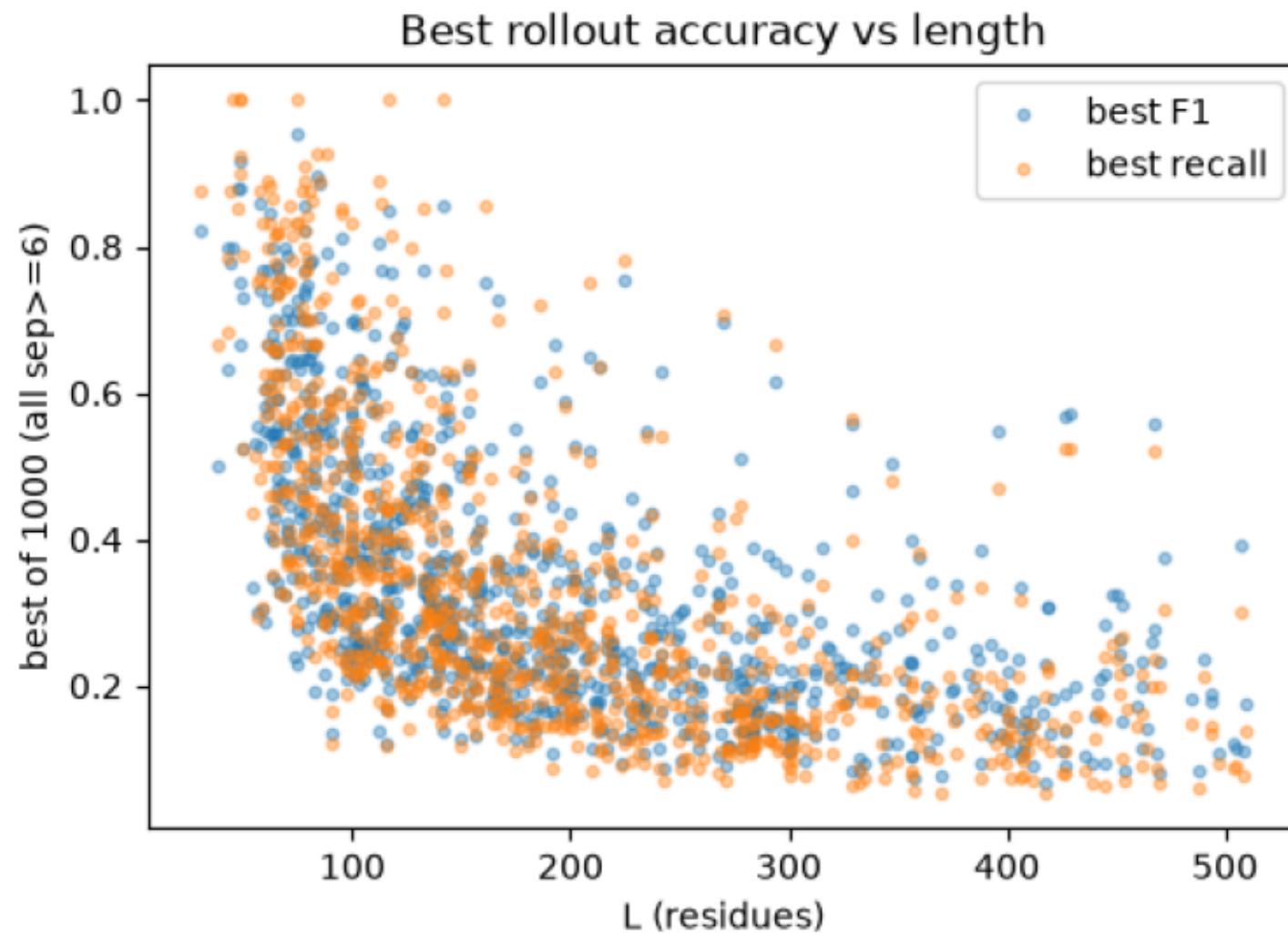
Best-of-1000 accuracy across 1000 targets (all sep $\geq$ 6)



```
$ aggregate_results.py --runs full=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full --out-prefix full
```

full_bestacc_vs_L.png

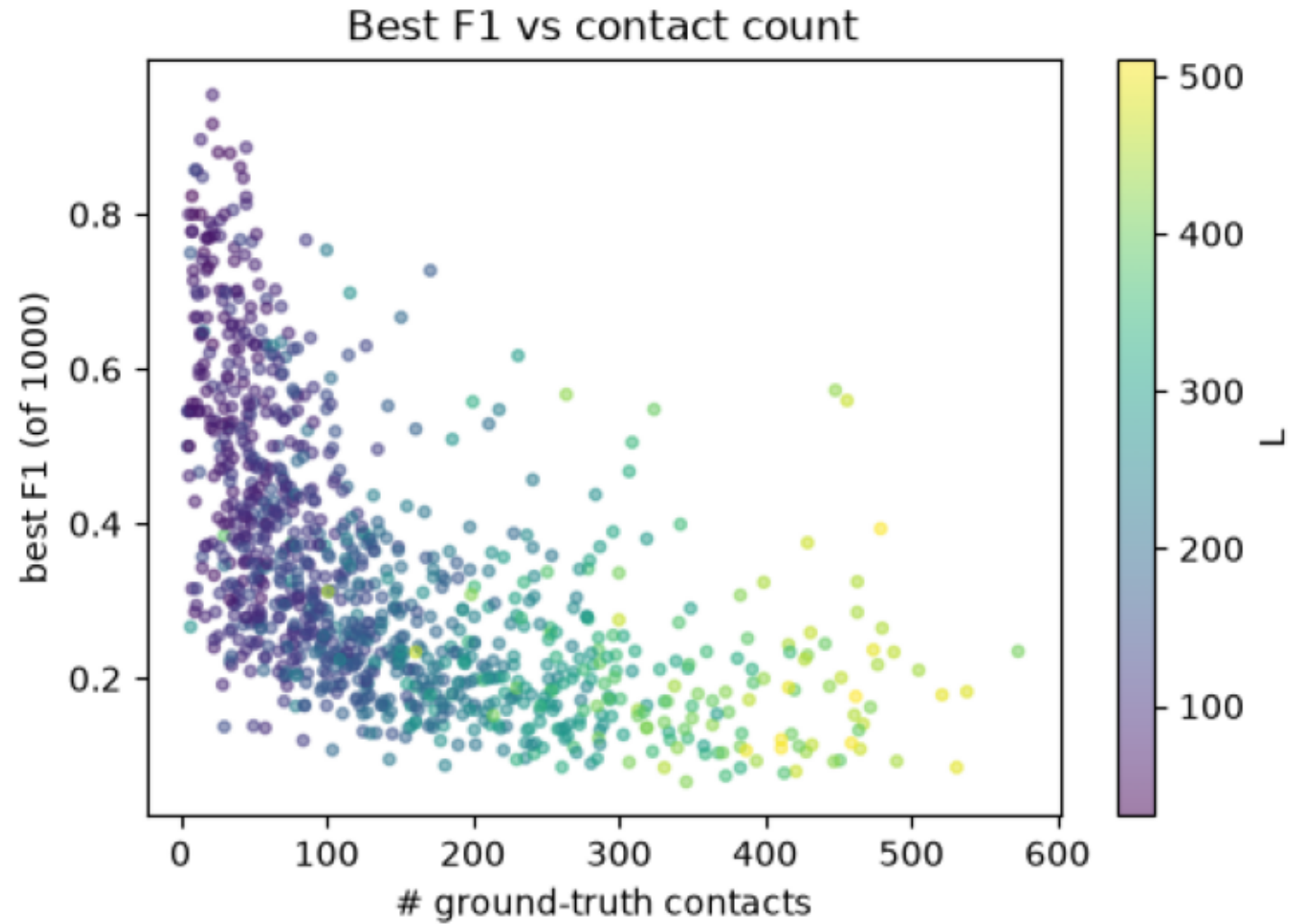
Per-target best-of-1000 recall/F1 vs protein length.



```
$ aggregate_results.py --runs full=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full --out-prefix full
```

full_bestf1_vs_ngt.png

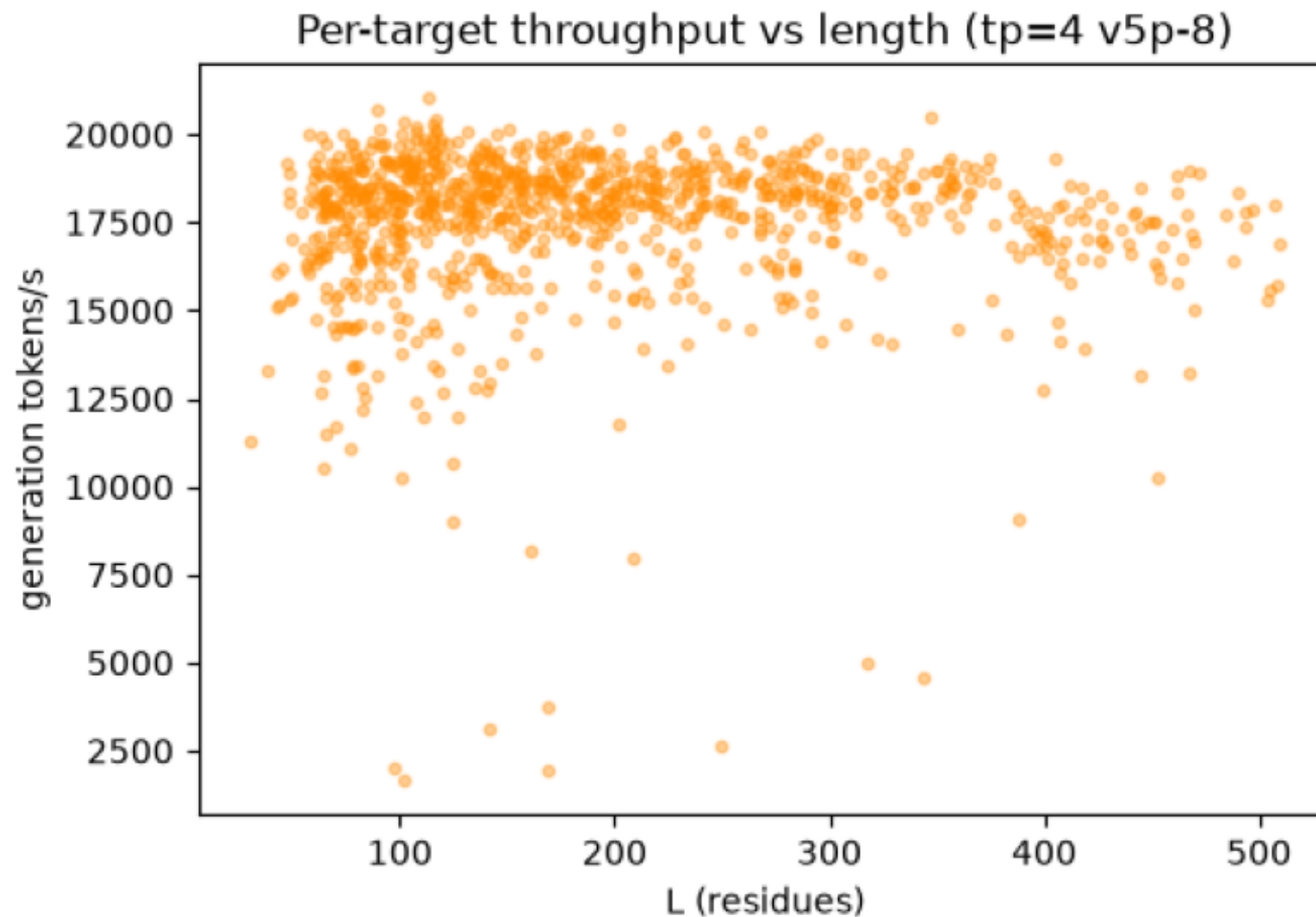
Per-target best-of-1000 F1 vs number of ground-truth contacts (colored by L).



```
$ aggregate_results.py --runs full=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full --out-prefix full
```


full_throughput_vs_L.png

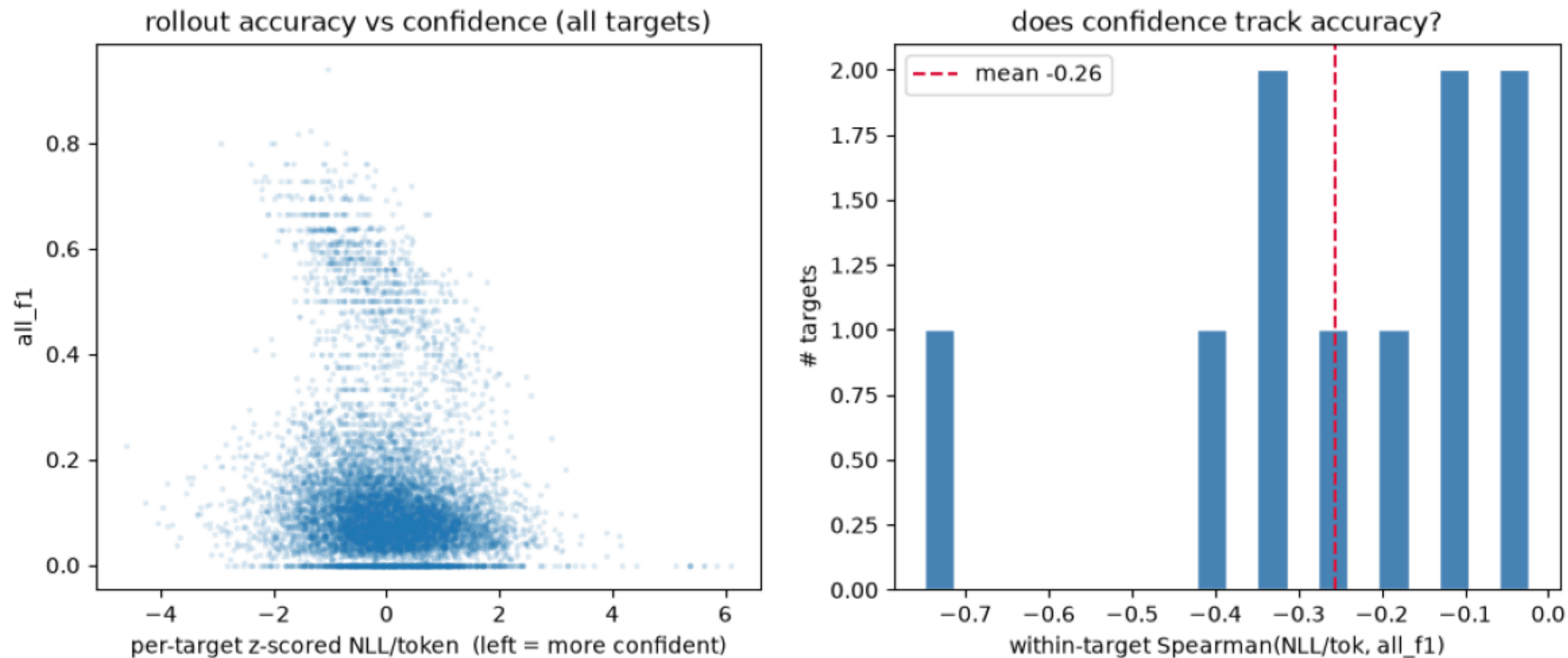
Per-target generation throughput (tokens/s) vs length.



```
$ aggregate_results.py --runs full=gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full --out-prefix full
```

nll_calib_f1_nll_vs_acc.png

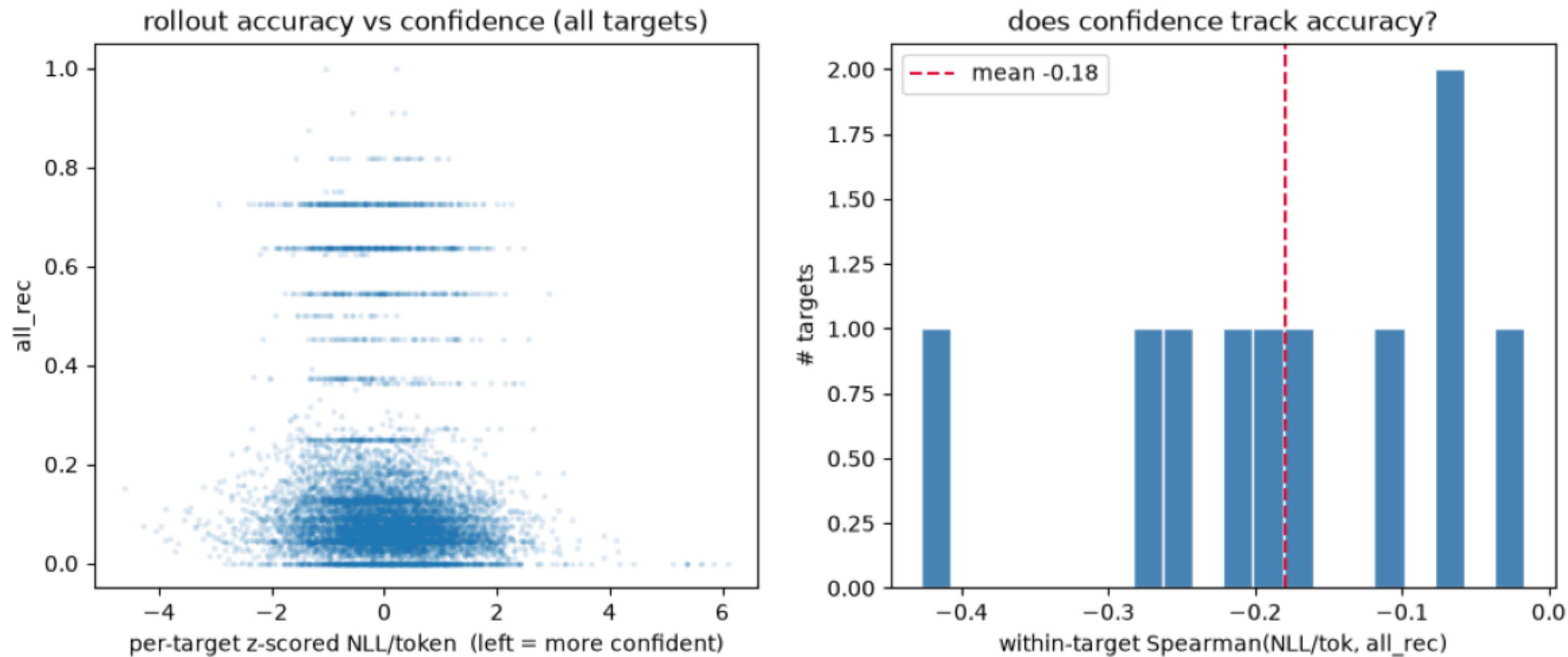
Left: rollout all_f1 vs per-target z-scored NLL/token. Right: per-target Spearman(NLL/tok, all_f1); negative = confident rollouts are more accurate.



```
$ analyze_nll.py --run gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/calib_nll --metric all_f1 --out-prefix nll_calib_f1
```

nll_calib_rec_nll_vs_acc.png

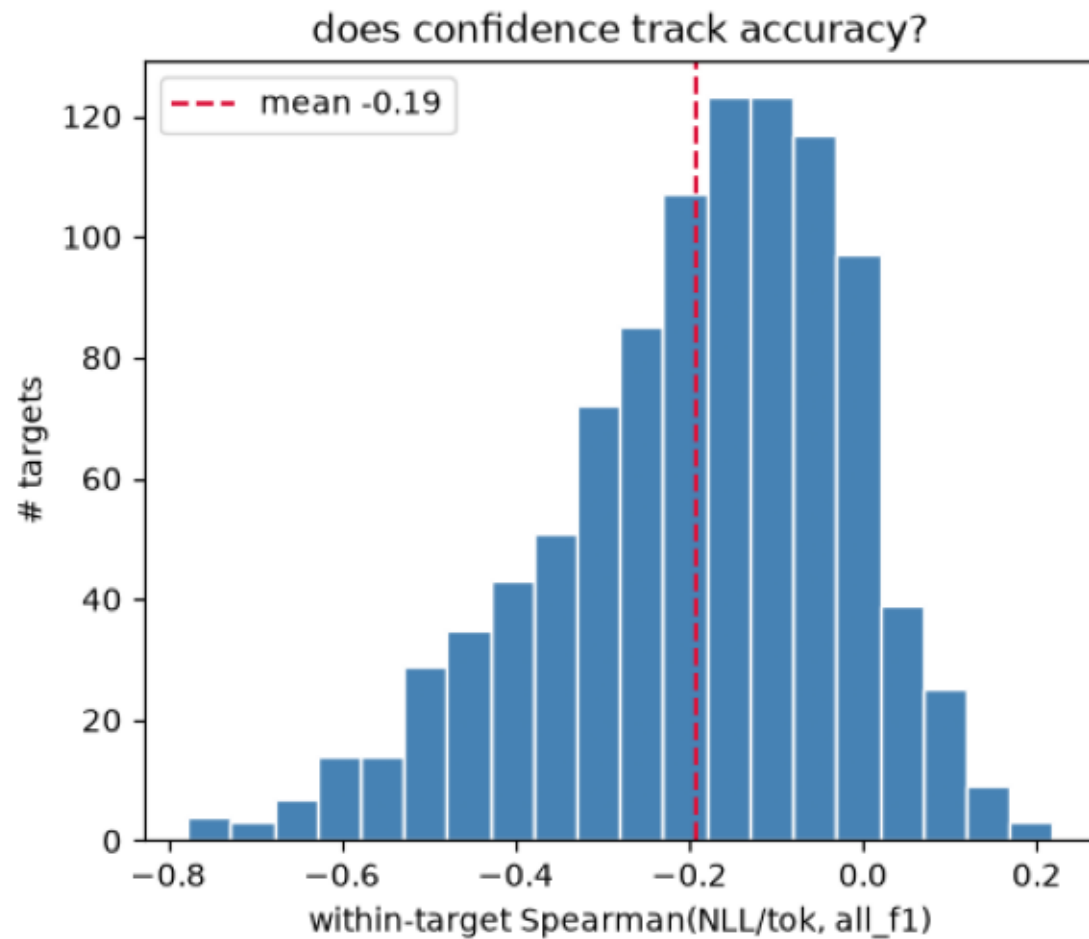
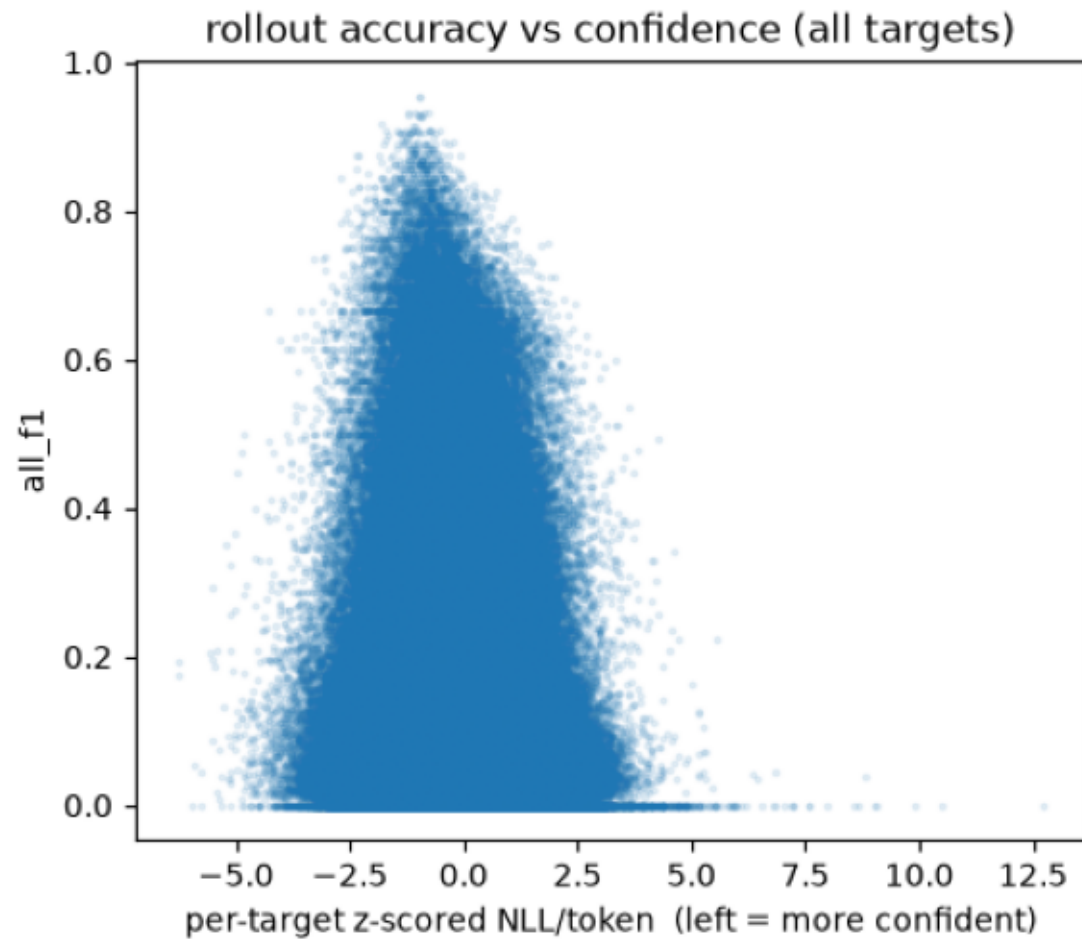
Left: rollout all_rec vs per-target z-scored NLL/token. Right: per-target Spearman(NLL/tok, all_rec); negative = confident rollouts are more accurate.



```
$ analyze_nll.py --run gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/calib_nll --metric all_rec --out-prefix nll_calib_rec
```

nll_full_f1_nll_vs_acc.png

Left: rollout all_f1 vs per-target z-scored NLL/token. Right: per-target Spearman(NLL/tok, all_f1); negative = confident rollouts are more accurate.



```
$ analyze_nll.py --run gs://marin-us-east5/protein-structure/MarinFold/exp98_rollouts_contacts_v1_train/runs/full_v2 --metric all_f1 --out-prefix nll_full_f1
```